Two Pointers

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Where you use two indices that traverse a data structure.
When to use two pointers:

- Sorted input
- Pairs or subarrays
- Sliding window problems
- Linked lists

Example: given a sorted array and a target, find id there exists any pair of elements, such that their sum is equal to the target.

1) Naïve method: time complexity O(n^2) and space O(1). It consists in generating all the possible pairs and check if any of them add up to the target value.
   # Function to check whether any pair exists
   # whose sum is equal to the given target value
```
def two_sum(arr, target):
    n = len(arr)

    # Iterate through each element in the array
    for i in range(n):

        # For each element arr[i], check every
        # other element arr[j] that comes after it
        for j in range(i + 1, n):

            # Check if the sum of the current pair
            # equals the target
            if arr[i] + arr[j] == target:
                return True

    # If no pair is found after checking
    # all possibilities
    return False

arr = [0, -1, 2, -3, 1]
target = -2

# Call the two_sum function and print the result
if two_sum(arr, target):
    print("true")
else:
    print("false")
```
2) Two pointes: time complexity O(n) and space is O(1)
   The idea is to begin with two corners of the given array. We use two index variables left and right to traverse from both corners. Run a loop while left < right.
   # Function to check whether any pair exists
   # whose sum is equal to the given target value
```
def two_sum(arr, target):
    # Sort the array

    left, right = 0, len(arr) - 1

    # Iterate while left pointer is less than right
    while left < right:
        sum = arr[left] + arr[right]
```

```
        # Check if the sum matches the target
        if sum == target:
            return True
        elif sum < target:
            left += 1  # Move left pointer to the right
        else:
            right -= 1 # Move right pointer to the left

    # If no pair is found
    return False

arr = [-3, -1, 0, 1, 2]
target = -2

# Call the two_sum function and print the result
if two_sum(arr, target):
    print("true")
else:
    print("false")
```